

Ingest (CPP-029)

CPP-Identifier	CPP-029
CPP-Label	Ingest
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Change history	Comments
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Version 1.1.0 - 2026-04-10	Migration to XML

1. Description of the CPP

The TDA performs all operations necessary to transform an SIP into an AIP.

Inputs and outputs

Input(s)	
Data	SIP
Metadata	Fixity metadata
	Descriptive metadata
	Provenance metadata (Optional)
	Technical metadata (Optional)
	Rights metadata (Optional)
Documentation/ guidance	Packaging policy
Output(s)	
Data	AIP
Metadata	Provenance metadata
	Technical metadata
	Rights metadata
Documentation/ guidance	Packaging policy

Definition and scope

Ingest is a high-level CPP, that is composed of or utilised by many other CPPs, and refers to the process of acquiring and incorporating data into a TDA. In its most abstract form, Ingest describes the process in which a TDA receives a *SIP* and transforms it into one or several *AIP(s)* through a process that includes **Data Quality Assessment** (CPP-019). At the end of a successful ingest, the data is preserved in the TDA for future use.

The Ingest process begins with the transfer of digital *Objects* and *Metadata* from their source environment to the TDA in the form of a *SIP*. The digital preservation workflow triggered by the submission ensures that the data is properly prepared for long-term storage, discovery and access. The workflow ensures that essential *Metadata* about the creation, structure, and context (CPP-016 **Metadata Ingest and Management**) of the data exists. *Fixity metadata* that will be essential for future preservation actions must also exist before the data can be preserved. In addition, *Technical metadata* that could not be easily extracted (e.g. column delimiters for CSV, quality assessment for OCR, or EPUB *Files* etc.) from the *Files* by the TDA may be required to be supplied by the producer.

Data Quality Assessment (CPP-019) measures are integrated throughout the ingest workflow to catch potential issues early. This includes **Virus Scanning** (CPP-007), **File Format Identification** (CPP-008), **Metadata Extraction** (CPP-009) and **File Format Validation** (CPP-010), and completeness checks to ensure that the digital *Objects* are suitable for preservation. The TDA verifies that transferred *Objects* are complete and uncorrupted through **Checksum Validation** (CPP-002). The process also involves assessing whether the digital *Objects* conform to the repository's technical requirements and collection development policies. Depending on the TDA's file format policy - preferred formats, *Objects* may also need to be normalised (CPP-026 **File Normalisation**) to preferred formats before or during the ingest process. During the ingest process, the TDA can generate additional *Metadata* and assign identifiers (CPP-005 Identifier Management) to support **Enabling Discovery** (CPP-024); perform **Enabling Access** (CPP-025) for cataloging purposes; and generate *Provenance metadata* that documents the transfer and processing history of the *Objects*. In order to evaluate the *SIP* and its contents, the TDA ensures that the *SIP* structure is valid and that the *SIP* is not incomplete (i.e. all *Objects* and *Metadata* are present).

SIPs that conform to the TDAs requirements and policies are transformed into *AIPs* which are sent to preservation in the archival storage of a TDA. *SIPs* that do not conform to the TDAs requirements will be handled according to its policies (in particular, file format policy and validation policy). The TDA can either reject the submitted data, ask the producer to address the issues before proceeding, flag the data as problematic and ingest it as it is, or perform an operation to address the identified issues.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
Submission of data to a TDA	

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓ sequence					
1	CPP-008 (File Format Identification) CPP-009 (Metadata Extraction) CPP-005 (Identifier Management)	Digital <i>Objects</i> <i>Metadata</i> provided by the producer	Pre-ingest actions (normalisation, metadata generation, identifier generation etc.) and <i>SIP</i> creation performed by the producer	<i>SIP</i>	CPP-005 (Identifier Management)
2		<i>SIP</i>	Submission of data to a TDA (<i>SIP</i> transfer)		
3		<i>SIP</i>	Identify whether the <i>SIP</i> is meant to create a new <i>AIP</i> or is intended to update one or several <i>AIPs</i> E.g. If the <i>SIP</i> has a Producer identifier that already corresponds to an <i>AIP</i> ingested in the system, proceed with the process	Updating request	CPP-021 (<i>AIP</i> Versioning)
4		<i>SIP</i>		Valid and complete <i>SIP</i>	

No	Supplier	Input	Steps	Output	Customer
		Packaging Policy	Ensure that the <i>SIP</i> structure conforms to the requirements and that its contents are not missing		
5	CPP-002 (Checksum Validation)	<i>Files</i> in the <i>SIP</i>	Perform checksum validation on each <i>File</i> in the <i>SIP</i>	<i>Information package</i> with fixity checked	
6	CPP-019 (Data Quality Assessment)	Quality assessment report	Quality assessment (ensuring that the submitted data conforms to requirements set by the TDA)	<i>Information package</i> with assessed quality	
		<i>SIP</i>		Result of the quality assessment recorded as <i>Provenance metadata</i>	
7	CPP-008 (File Format Identification) CPP-009 (Metadata Extraction) CPP-007 (Virus Scanning)	<i>SIP</i>	Perform <i>File</i> Format Identification, <i>Metadata</i> Extraction, and Virus Scanning	<i>Technical metadata</i>	
				<i>Provenance metadata</i>	
8	CPP-010 (File Format Validation)	Format policy - Validation <i>Files</i> in the <i>SIP</i>	<i>Optional:</i> Perform Format Validation if the TDAs format policy states that validation must be performed	<i>Technical metadata</i>	
				<i>Provenance metadata</i>	
9	CPP-020 (Rights Management)	<i>Objects</i> in the <i>SIP</i> Rights assessment	Perform rights assessment on <i>Objects</i> contained in the <i>SIP</i>	<i>Rights metadata</i>	

No	Supplier	Input	Steps	Output	Customer
10	CPP-016 (Metadata Ingest and Management)	<i>Metadata</i> provided by the producer <i>Technical metadata</i> <i>Provenance metadata</i> <i>Rights metadata</i>	Record the <i>Metadata</i> provided by the producer and produced by the TDA according to the TDAs <i>SIP</i> requirements and policy of automatic enrichment of <i>SIP Metadata</i>	<i>Information package</i> with <i>Metadata</i> recorded	
11	CPP-005 (Identifier Management)	Identifier	Assign identifier to the <i>Information package</i>	<i>Information package</i> with Identifier assigned	
12	CPP-026 (File Normalisation)	<i>Files</i> in the <i>SIP</i>	<i>Optional:</i> Only if the TDA supports normalisation during ingest: Normalisation of data, including documenting the actions	New <i>Representations</i> in a supported format <i>Provenance metadata</i>	
13	CPP-028 (Creation of Derivatives)	<i>Files</i> in the <i>SIP</i>	<i>Optional:</i> Only if the TDA supports creating derivatives during ingest: Generation of derivatives	New additional <i>Representations</i>	
14	↔ <i>alternative</i>				
	14.a	↓ <i>sequence</i>			

No			Supplier	Input	Steps	Output	Customer
		14.a.1		<i>Information package</i> with fixity checked, identifier assigned, quality assessed, <i>Metadata</i> recorded and optionally new <i>Representations</i> added	If <i>SIP</i> conforms to the requirements (steps 7a and 7b): <i>SIP</i> transformation to <i>AIP</i>	<i>AIP</i>	
		14.a.2		<i>AIP</i>	Move the <i>AIP</i> to the archival storage	<i>AIP</i> on multiple locations	CPP-011 (Replication)
	14.b			Error-handling policies (in particular file format policy and validation policy)	If <i>SIP</i> doesn't conform to the requirements, perform one of these actions: (a) Rejection of <i>SIP</i> ; (b) Request the producer to address the issues; (c) Flag the data as problematic and ingest it as it is; (d) Perform an operation to address the identified issues.	Error report to the submitter of the <i>SIP</i> (producer)	
						Trigger technical analysis	
15					Notification / report to the producer about the outcome of the ingest	Ingest report	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
Ingest transfers the responsibility from the creator/owner/depositor of the digital <i>Objects</i> to the TDA, enabling long-term preservation, discovery and access to the digital <i>Objects</i>. Also, the ingest process captures the <i>Objects</i>' state at the time of transfer through checksums, metadata extraction, and documentation of the transfer process itself. This creates an auditable trail that supports future authenticity claims and helps detect any corruption or unauthorised modifications that may occur over time.	The digital <i>Objects</i> remain vulnerable to loss, corruption, or unauthorised changes in their original environment. Furthermore, <i>Objects</i> in their original environments are often stored in formats, structures, or contexts that are not optimal for long-term preservation.

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
CPP-002	Checksum Validation	Must be performed during ingest.
CPP-005	Identifier Management	Must be performed during ingest.
CPP-007	Virus Scanning	Must be performed during ingest.
CPP-008	File Format Identification	Must be performed during ingest.
CPP-009	Metadata Extraction	Must be performed during ingest.
CPP-020	Rights Management	Some minimal rights assessment must be performed during ingest to verify that the TDA should be in charge of preserving the content of the <i>SIP</i> .
CPP-016	Metadata Ingest and Management	The ingest process produces <i>Technical, Rights and Provenance metadata</i> that are recorded in the <i>Information package</i> and digital archive database by <i>Metadata Ingest and Management</i> .

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
May require	CPP-010	File Format Validation	A TDA may validate the format of the submitted <i>Files</i> in the ingest phase.
May require	CPP-019	Data Quality Assessment	The TDA may have quality requirements that may be checked during ingest.
May require	CPP-026	File Normalisation	The ingest may require that the digital <i>Objects</i> are first normalised before ingestion.
Required by	CPP-021	AIP Versioning	Versioning implies several delicate operations, in particular in the case of a partial update, where the incoming <i>SIP</i> should be merged with the existing <i>AIP</i> .
May be required by	CPP-028	Creation of Derivatives	The ingestion may generate derivatives for access.
Affinity with	CPP-012	Risk Mitigation	The ingest process must adhere to the risk mitigation policies.
Affinity with	CPP-013	Object Management Reporting	Ingest is both an important provider of reporting data to the TDA (via other CPPs) as well as a customer, as the ingest checks and outcomes must be reported to the producer.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	Ingest	/
Nestor Seal	Ingest	C14 Integrity: Ingest Interface C17 Authenticity: Ingest
ISO 16363	Ingest	4.1. Ingest: Acquisition of content 4.2. Ingest: Creation of the <i>AIP</i>

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS	Ingest	4.2.2. General - Figure 4-1
	Ingest Functional Entity	4.2.3.3. Ingest
PREMIS	Ingest	Glossary

4. Reference implementations

Use cases

Ingest processes described as workflows

Institutional background	
Institution	Several institutions, MUL
Hyperlink	https://coptr.digipres.org/index.php/Workflow:Community_Owned_Workflows
Description	
Problem statement	Several institutions have described their workflow for ingesting <i>Objects</i> in the Community Owned Workflow section of the COPTR registry.

Publicly available documentation

Institution	Organisation type	Language	Hyperlink
TIB – Leibniz Information Centre for Science and Technology and University Library, DE	National library; Non-commercial digital preservation service; Research infrastructure; Research performing organisation	English	https://wiki.tib.eu/confluence/spaces/lza/pages/93608618/Ingest
CSC – IT Center for Science Ltd., FI	Non-commercial digital preservation service	Finnish	https://urn.fi/urn:nbn:fi-fe2024051731943 (Appendix 4, section 2.2.1)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/transfer/transfer/ (Transfer - steps that lead up to creating a SIP, e.g. checksum generation, file format identification etc.)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/ingest/ingest/ (Ingest - steps from SIP to AIP, e.g. file format normalisation)

